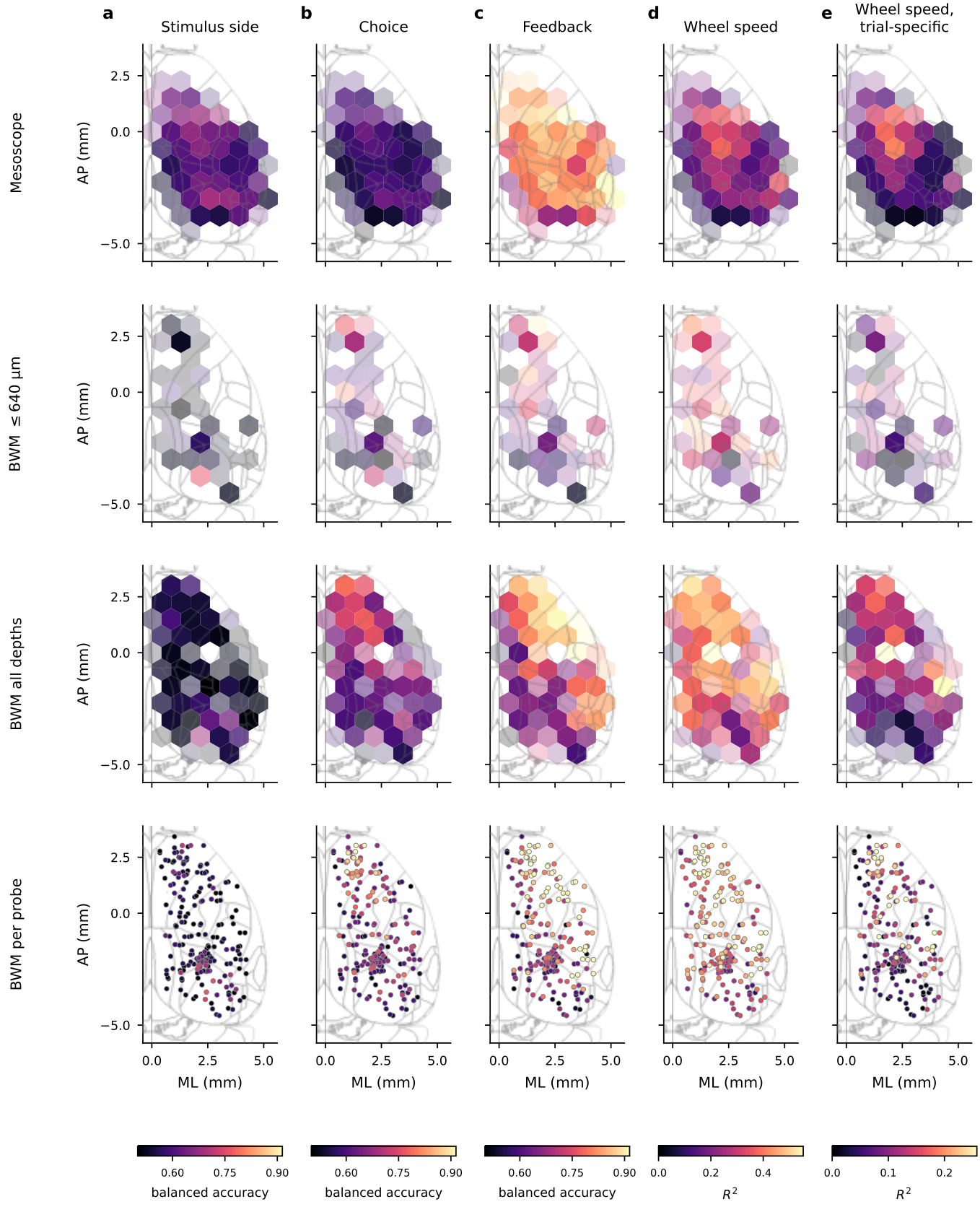


Figure 1: Decoding from all neurons. Rows: mesoscope; BWM units within 640 μm of the pial surface; BWM at all depths. Each bin (1 mm hexagon, hemispheres folded) shows the mean over sessions of decoders trained on all neurons of one session in that bin (sessions with $\geq$20 neurons there); mesoscope bins hold far more neurons than BWM bins. Opacity: number of animals (full at $\geq$4). Bottom row: BWM decoded per probe from all its isocortex units at all depths, whichever bins they fall in (209 probes with $\geq$20 units, median 40); each circle sits at the probe's brain entry point (histology-aligned Alyx trajectory, hemispheres folded), higher scores drawn on top where circles overlap; colour scales as in the rows above, so circles beyond the top of a scale saturate. **a** Stimulus side, balanced accuracy (chance 0.5; colour scale shared by **a**–**c**). **b** Choice. **c** Feedback. **d** Wheel speed, cross-validated R^2. **e** Trial-specific wheel speed, R^2.